

Newton's Second Law

Resultant force = ma , in the direction of the acceleration. Connected particles: one equation per particle, or one for the whole system.



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THE TOPIC ON ONE MAP

Newton's Second Law

01 One body

$F = ma$ lifts falling with resistance

02 Connected in a line

towing whole system, then one part

03 Pulleys: hanging, table

one equation each force on the pulley string breaks

04 Pulleys with a slope

friction $\tan \alpha = \frac{3}{4}$ direction of **S**

NEWTON'S SECOND LAW

(forces WITH the acceleration) – (forces AGAINST it) = ma

in equilibrium or at constant speed $a = 0$

“light, inextensible string”: same

tension, same a

01 One Body: Resultant Force = ma

Pick the direction of the acceleration as positive. Add the forces that way, subtract the forces the other way, and set it equal to ma . Then SUVAT if a speed, time or distance is wanted.

WHEN YOU SEE → YOU DO

“driving force ... resistance”

→ $D - R = ma$

a lift or a bucket moving up / down

→ $T - mg = ma$ (up) — use the direction of a

“the force on the man from the floor”

→ treat the man alone: $R - mg = ma$

“falls against a constant resistance”

→ $mg - R = ma$

a rough horizontal surface

→ friction μR against the motion

a collision, then it slides

→ momentum first, then $F = ma$ with friction

WORKED MODEL • OCTOBER 2022

Lift 250 kg; cable tension 3616 N; the children push down on the floor with $565 + 226$ N. Find a .

1 $3616 - 250g - 791 = 250a$

forces on the lift only

2 $375 = 250a$

ANSWER $a = 1.5 \text{ m s}^{-2}$

× **TRAP** Using $F = ma$ with the forces on the *children* when the question says “forces acting on the lift only”.

IDEA 01

Practise: One Body: Resultant Force = ma

My questions first, then real past papers.

Q1 DR ESLAM Driving force and resistance

M1 • TOPIC 7 2 marks

A car of mass 1200 kg on a level road has driving force 3000 N and resistance 600 N. Find its acceleration.

Q2 DR ESLAM A lift

M1 • TOPIC 7 4 marks

A lift of mass 500 kg carries a man of mass 60 kg and accelerates upwards at 0.5 m s^{-2} . Find the tension in the cable and the reaction on the man.

Whole system for T ; the man alone for R .

Q3 **DR ESLAM** Falling with resistance**M1 · TOPIC 7** **3 marks**

A stone of mass 0.5 kg falls from rest against a constant resistance of 1.4 N. Find its acceleration and its speed after 3 s.

PAST PAPERS

Q4 PAST PAPER Pearson IAL WME01 • June 2021 • Q4

M1 • TOPIC 7 6 marks

4.

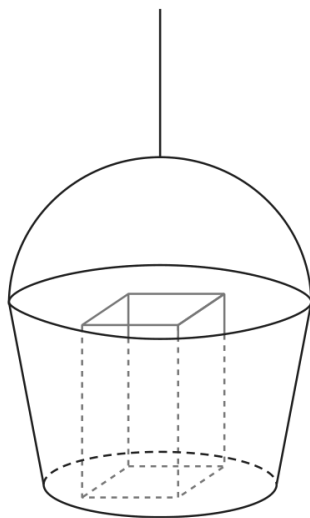
**Figure 1**

Figure 1 shows a large bucket used by a crane on a building site to move materials between the ground and the top of the building. The mass of the bucket is 15 kg.

The bucket is attached to a vertical cable with the bottom of the bucket horizontal. The cable is modelled as light and inextensible.

When the bucket is on the ground, a bag of cement of mass 25 kg is placed in the bucket.

The bucket with the bag of cement moves vertically upwards with constant acceleration 0.2 ms^{-2} . Air resistance is modelled as being negligible.

(a) Find the tension in the cable.

(3)

At the top of the building, the bag of cement is removed. A box of tools of mass 12 kg is now placed in the bucket.

Later on the bucket with the box of tools is moving vertically downwards with constant deceleration 0.1 ms^{-2} . Air resistance is again modelled as being negligible.

(b) Find the magnitude of the normal reaction between the bucket and the box of tools.

(3)

Q5**PAST PAPER**

Pearson IAL WME01 · October 2021 · Q2

M1 · TOPIC 7 10 marks

2. A particle P of mass $2m$ is moving on a rough horizontal plane when it collides directly with a particle Q of mass $4m$ which is at rest on the plane. The speed of P immediately before the collision is $3u$. The speed of Q immediately after the collision is $2u$.

(a) Find, in terms of u , the speed of P immediately after the collision.

(3)

(b) State clearly the direction of motion of P immediately after the collision.

(1)

Following the collision, Q comes to rest after travelling a distance $\frac{6u^2}{g}$ along the plane.

The coefficient of friction between Q and the plane is μ .

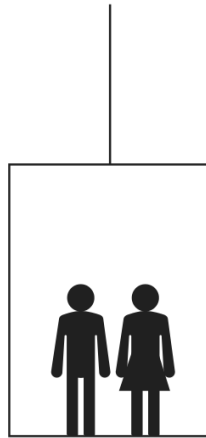
(c) Find the value of μ .

(6)

Q6 PAST PAPER Pearson IAL WME01 • October 2022 • Q4

M1 • TOPIC 7 6 marks

4.

**Figure 3**

Two children, Alan and Bhavana, are standing on the horizontal floor of a lift, as shown in Figure 3.

The lift has mass 250 kg. The lift is raised vertically upwards with constant acceleration by a vertical cable which is attached to the top of the lift. The cable is modelled as being light and inextensible. While the lift is accelerating upwards, the tension in the cable is 3616 N.

As the lift accelerates upwards, the floor of the lift exerts a force of magnitude 565 N on Alan and a force of magnitude 226 N on Bhavana.

Air resistance is modelled as being negligible and Alan and Bhavana are modelled as particles.

(a) By considering the forces acting on the lift only, find the acceleration of the lift.

(3)

(b) Find the mass of Alan.

(3)

Q7 PAST PAPER Pearson IAL WME01 · June 2023 · Q3

M1 · TOPIC 7 8 marks

3. Two students observe a book of mass 0.2 kg fall vertically from rest from a shelf that is 1.5 m above the floor.

Student A suggests that the book is modelled as a particle falling freely under gravity.

- (a) Use student A 's model to find the time taken for the book to reach the floor.

(3)

Student B suggests an improved model where the book is modelled as a particle experiencing a constant resistance to motion of magnitude R newtons.

Given that the time taken for the book to reach the floor is 0.6 seconds,

- (b) use student B 's model to find the value of R

(5)

Q8 PAST PAPER Pearson IAL WME01 · October 2023 · Q3

M1 · TOPIC 7 10 marks

3. A hammer is used to hit a tent peg into soft ground.

The hammer has mass 1.8 kg and the tent peg has mass 0.2 kg .

The hammer and tent peg are both modelled as particles and the impact is modelled as a direct collision.

Immediately before the impact, the tent peg is stationary and the hammer is moving vertically downwards with speed 10 m s^{-1}

Immediately after the impact, the hammer and tent peg move together, vertically downwards, with the **same** speed $v\text{ m s}^{-1}$

- (a) Find the value of v (2)

- (b) Find the magnitude of the impulse exerted on the tent peg by the hammer, stating the units of your answer. (3)

The ground exerts a constant vertical resistive force of magnitude R newtons, bringing the hammer and tent peg to rest after they travel a distance of 12 cm .

- (c) Find the value of R . (5)

Q9 PAST PAPER Pearson IAL WME01 • January 2025 • Q3

M1 • TOPIC 7 12 marks

3.

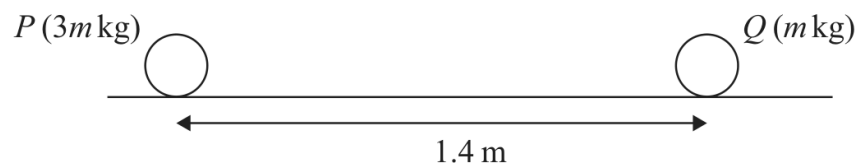


Figure 1

A particle P of mass $3m \text{ kg}$ and a particle Q of mass $m \text{ kg}$ are at rest on a rough horizontal surface. The distance between P and Q is 1.4 m , as shown in Figure 1.

An impulse of magnitude $\lambda \text{ N s}$ is now applied to P in the direction PQ . Immediately after the impulse is applied, the speed of P is 5 m s^{-1} .

(a) Find λ in terms of m .

(2)

Immediately before P collides with Q , the speed of P is 2.5 m s^{-1} . The coefficient of friction between P and the surface is μ .

(b) Find the value of μ .

(7)

Immediately after P collides with Q , the speed of Q is 2.1 m s^{-1} .

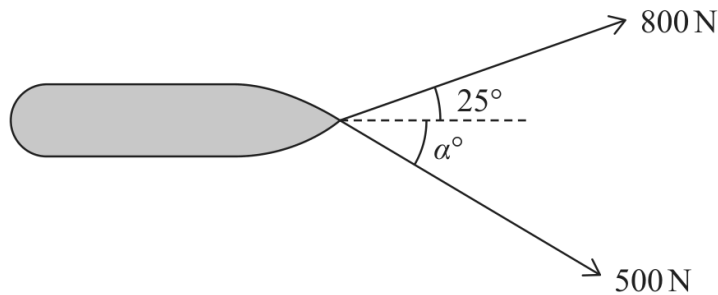
(c) Find the speed of P immediately after the collision.

(3)

Q10 PAST PAPER Pearson IAL WME01 · June 2025 · Q4

M1 · TOPIC 7 7 marks

4.

**Figure 2**

Two ropes are attached to a point on the front of a barge.
The barge is being pulled horizontally in a straight line along the centre of a long straight canal.

One rope makes an angle of 25° with the direction of motion of the barge and has a tension of 800 N.

The other rope makes an angle of α° with the direction of motion of the barge and has a tension of 500 N, as shown in Figure 2.

Both ropes are horizontal.

(a) Find the value of α

(3)

The mass of the barge is 15 tonnes and the resistance to the motion of the barge is a constant force of magnitude 750 N.

(b) Find the acceleration of the barge.

(4)

02 Connected in a Line: Towing and Lifts

Everything moves with the same acceleration. First treat the whole system as one body (the towbar or string tension is internal and cancels) to find a . Then take **one part alone** to find the tension.

WHEN YOU SEE → YOU DO

a car towing a trailer

→ whole: $D - R_1 - R_2 = (m_1 + m_2)a$; trailer: $T - R_2 = m_2a$

“up a road inclined at α ”

→ add $mg \sin \alpha$ against the motion

“the towbar is modelled as a light rod”

→ same tension along it; it can push (thrust)

boxes stacked in a lift

→ whole for the cable; top box alone for the contact force

“state how you used ... inextensible”

→ both parts have the same acceleration

Whole system for a

One part for T

Check with the other part

WORKED MODEL • JUNE 2023

Car 1200 kg tows trailer 600 kg up $\sin \alpha = \frac{1}{12}$; $D = 3000$ N, resistances $2R$ and R , $a = 0.75$. Show $R = 60$.

$$1 \quad 3000 - 3R - 1800g \cdot \frac{1}{12} = 1800(0.75)$$

whole system

$$2 \quad 3000 - 3R - 1470 = 1350$$

ANSWER $R = 60$

IDEA 02

Practise: Connected in a Line: Towing and Lifts

My questions first, then real past papers.

Q11 DR ESLAM Car and trailer

M1 · TOPIC 7 4 marks

A car (1000 kg) tows a trailer (500 kg) on a level road. Driving force 2400 N; resistances 300 N (car) and 150 N (trailer). Find the acceleration and the tension in the towbar.

Whole system for a ; then the trailer alone for T .

Q12 DR ESLAM Two particles on one string

M1 · TOPIC 7 4 marks

A force of 60 N pulls a 2 kg particle vertically upwards; a 3 kg particle hangs below it on a light string. Find the acceleration and the tension in the string.

PAST PAPERS

Q13 PAST PAPER Pearson IAL WME01 · October 2019 · Q3

M1 · TOPIC 7 7 marks

3. A car of mass 800 kg is towing a trailer of mass 400 kg up a straight road using a towbar. The towbar is parallel to the road and parallel to the direction of motion of the car. The road is inclined to the horizontal at an angle α , where $\sin \alpha = \frac{1}{7}$. The engine of the car produces a constant driving force of magnitude D newtons. The resistance to the motion of the car from non-gravitational forces is modelled as a single force of magnitude 420 N. The resistance to the motion of the trailer from non-gravitational forces is modelled as a single force of magnitude 300 N. The car and trailer are modelled as particles and the towbar is modelled as a light rod.

Given that the tension in the towbar is 2060 N, find the value of D .

(7)

Q14 PAST PAPER Pearson IAL WME01 · June 2022 · Q3

M1 · TOPIC 7 7 marks

3. A tractor of mass 6 tonnes is dragging a large block of mass 2 tonnes along rough horizontal ground. The cable connecting the tractor to the block is horizontal and parallel to the direction of motion.

The cable is modelled as being light and inextensible.

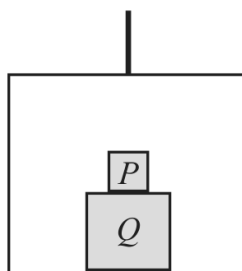
The driving force of the tractor is 7400 N and the resistance to the motion of the tractor is 200 N. The resistance to the motion of the block is R newtons, where R is a constant.

Given that the tension in the cable is 6000 N and the tractor is accelerating,

- (a) find the value of R . (6)
- (b) State how you have used the fact that the cable is modelled as being inextensible. (1)

Q15 PAST PAPER Pearson IAL WME01 · January 2023 · Q7**M1 · TOPIC 7 7 marks**

7.

**Figure 4**

A simple lift operates by means of a vertical cable which is attached to the top of the lift.

The lift has mass m

A box Q is placed on the floor of the lift.

A box P is placed directly on top of box Q , as shown in Figure 4.

The cable is modelled as being light and inextensible and air resistance is modelled as being negligible.

The tension in the cable is $\frac{42mg}{5}$

The lift and its contents move vertically upwards with acceleration $\frac{2g}{5}$

Using the model,

(a) find, in terms of m , the combined mass of boxes P and Q (4)

During the motion of the lift, the force exerted on box P by box Q is $\frac{14mg}{5}$

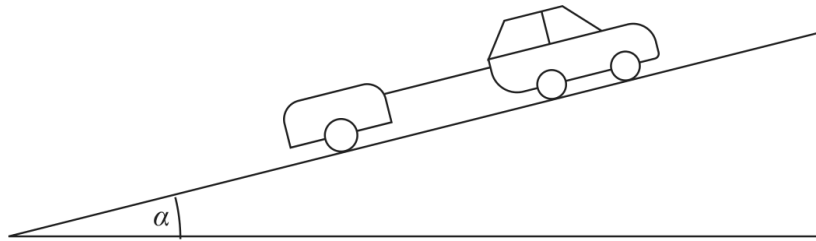
Using the model,

(b) find, in terms of m , the mass of box P (3)

Q16 PAST PAPER Pearson IAL WME01 · June 2023 · Q7

M1 · TOPIC 7 11 marks

7.

**Figure 4**

A car of mass 1200 kg is towing a trailer of mass 600 kg up a straight road, as shown in Figure 4.

The road is inclined at an angle α to the horizontal, where $\sin \alpha = \frac{1}{12}$

The driving force produced by the engine of the car is 3000 N.

The car moves with acceleration 0.75 m s^{-2}

The non-gravitational resistance to motion of

- the **car** is modelled as a constant force of magnitude $2R$ newtons
- the **trailer** is modelled as a constant force of magnitude R newtons

The car and the trailer are modelled as particles.

The tow bar between the car and trailer is modelled as a light rod that is parallel to the direction of motion.

Using the model,

(a) show that the value of R is 60 (4)

(b) find the tension in the tow bar. (3)

When the car and trailer are moving at a speed of 12 m s^{-1} , the tow bar breaks.

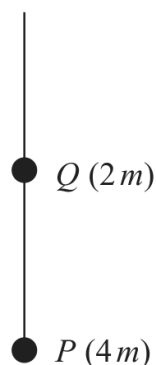
Given that the non-gravitational resistance to motion of the trailer remains unchanged,

(c) use the model to find the further distance moved by the trailer before it first comes to rest. (4)

Q17 PAST PAPER Pearson IAL WME01 · June 2024 · Q3

M1 · TOPIC 7 6 marks

3.

**Figure 1**

Two particles, P and Q , have masses $4m$ and $2m$ respectively. The particles are connected by a light inextensible string. A second light inextensible string has one end attached to Q . Both strings are taut and vertical, as shown in Figure 1.

The particles are **accelerating** vertically **downwards**.

Given that the tension in the string connecting the two particles is $3mg$, find, in terms of m and g , the tension in the upper string.

(6)

03 Pulleys: Hanging and on a Table

Over a smooth pulley the tension is the same on both sides and both particles have the same size of acceleration. Write one equation for each particle, each in its **own** direction of motion, and add them to lose T .

WHEN YOU SEE → YOU DO

two particles hanging over a pulley

→ heavy: $m_1g - T = m_1a$; light: $T - m_2g = m_2a$

one on a table, one hanging over the edge

→ table: $T - F = m_1a$; hanging: $m_2g - T = m_2a$

“force exerted on the pulley”

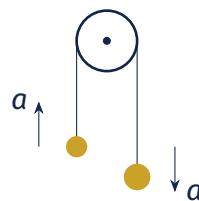
→ two tensions: $2T$ (vertical strings) or $T\sqrt{2}$ (at 90°)

“B hits the floor and does not rebound”

→ string slack: A now moves alone, new a

“the string breaks”

→ each particle moves freely under gravity



light: $T - m_2g = m_2a$

heavy: $m_1g - T = m_1a$

add: $(m_1 - m_2)g = (m_1 + m_2)a$

WORKED MODEL • OCTOBER 2024

P (3 kg) and Q (5 kg) hang over a smooth pulley. Find T and the force on the pulley.

1 $5g - T = 5a, T - 3g = 3a \Rightarrow a = \frac{g}{4}$

add

2 $T = 3(g + \frac{g}{4}) = \frac{15g}{4}$

ANSWER $T \approx 36.8 \text{ N}, F = 2T \approx 73.5 \text{ N}$

IDEA 03

Practise: Pulleys: Hanging and on a Table

My questions first, then real past papers.

Q18 DR ESLAM Two hanging masses

M1 • TOPIC 7 4 marks

Particles of 3 kg and 5 kg hang over a smooth pulley and are released from rest. Find the acceleration and the tension.

One equation for each particle, each in its own direction of motion.

Q19 DR ESLAM Table and edge

M1 • TOPIC 7 4 marks

A 2 kg block on a smooth table is joined over a pulley at the edge to a 1 kg particle hanging freely. Find the acceleration, the tension and the force on the pulley.

PAST PAPERS

Q20 PAST PAPER Pearson IAL WME01 · January 2020 · Q7

M1 · TOPIC 7 18 marks

7.

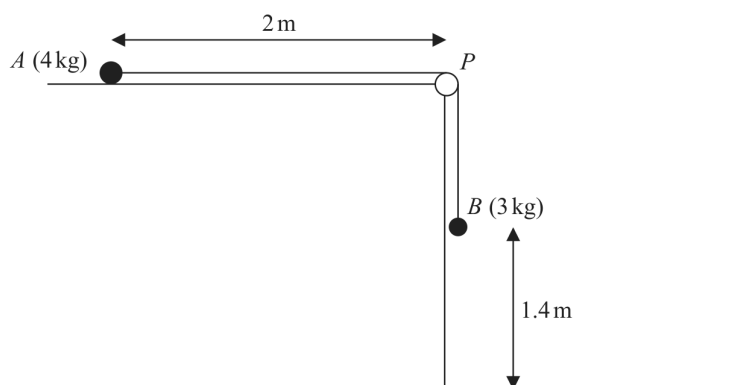


Figure 3

A particle A of mass 4 kg is held at rest on a rough horizontal table. Particle A is attached to one end of a string that passes over a pulley P . The pulley is fixed at the edge of the table. The other end of the string is attached to a particle B , of mass 3 kg , which hangs freely below P .

The part of the string from A to P is perpendicular to the edge of the table and A , P and B all lie in the same vertical plane.

The string is modelled as being light and inextensible and the pulley is modelled as being small, smooth and light.

The system is released from rest with the string taut. At the instant of release, A is 2 m from the edge of the table and B is 1.4 m above a horizontal floor, as shown in Figure 3.

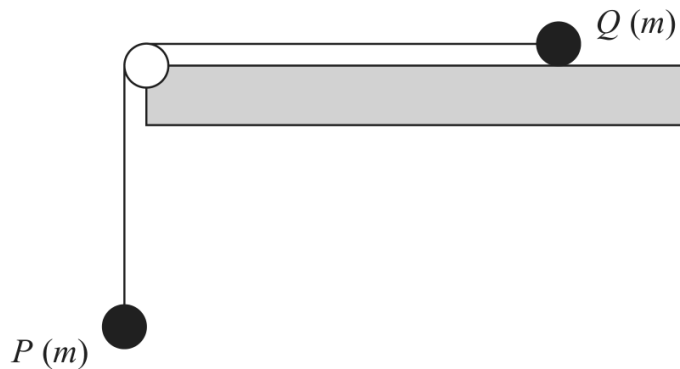
After descending with constant acceleration for 2 seconds , B hits the floor and does not rebound.

- (a) Show that the acceleration of A before B hits the floor is 0.7 m s^{-2} (2)
 - (b) State which of the modelling assumptions you have used in order to answer part (a). (1)
 - (c) Find the magnitude of the resultant force exerted on the pulley by the string. (4)
- The coefficient of friction between A and the table is μ .
- (d) Find the value of μ . (6)
 - (e) Determine, by calculation, whether or not A reaches the pulley. (5)

(Total 18 marks)

Q21 PAST PAPER Pearson IAL WME01 · October 2022 · Q7**M1 · TOPIC 7** 13 marks

7.

**Figure 4**

A particle P of mass m is attached to one end of a light inextensible string. Another particle Q , also of mass m , is attached to the other end of the string. The string passes over a small smooth pulley which is fixed at the edge of a rough horizontal table. Particle Q is held at rest on the table and particle P hangs vertically below the pulley with the string taut, as shown in Figure 4.

The pulley, P and Q all lie in the same vertical plane.

The coefficient of friction between Q and the table is μ , where $\mu < 1$

Particle Q is released from rest.

The tension in the string before Q hits the pulley is kmg , where k is a constant.

(a) Find k in terms of μ .

(7)

Given that Q is initially a distance d from the pulley,

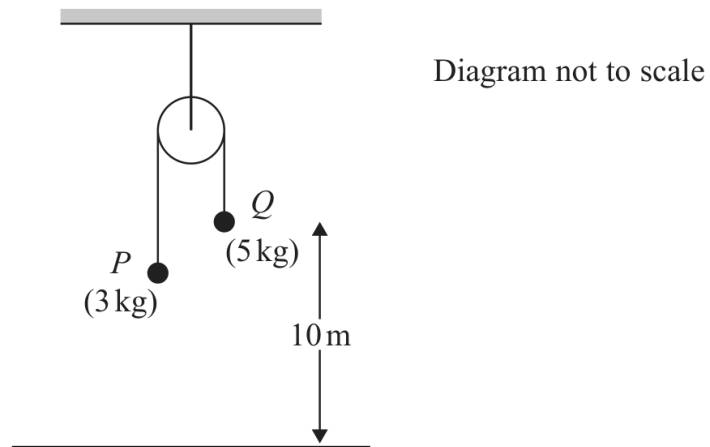
(b) find, in terms of d , g and μ , the time taken by Q , after release, to reach the pulley.

(4)

(c) Describe what would happen if $\mu \geq 1$, giving a reason for your answer.

(2)

5.

**Figure 3**

Two particles, P and Q , have masses 3 kg and 5 kg respectively. The particles are connected by a light inextensible string which passes over a small smooth fixed pulley.

The particles are released from rest with the string taut and the hanging parts of the string vertical, as shown in Figure 3.

Immediately after the particles are released from rest, P moves upwards with acceleration $a\text{ ms}^{-2}$ and the tension in the string is T newtons.

(a) Write down an equation of motion for P . (2)

(b) Find the value of T . (4)

The total force acting on the pulley due to the string has magnitude F newtons.

(c) Find the value of F . (2)

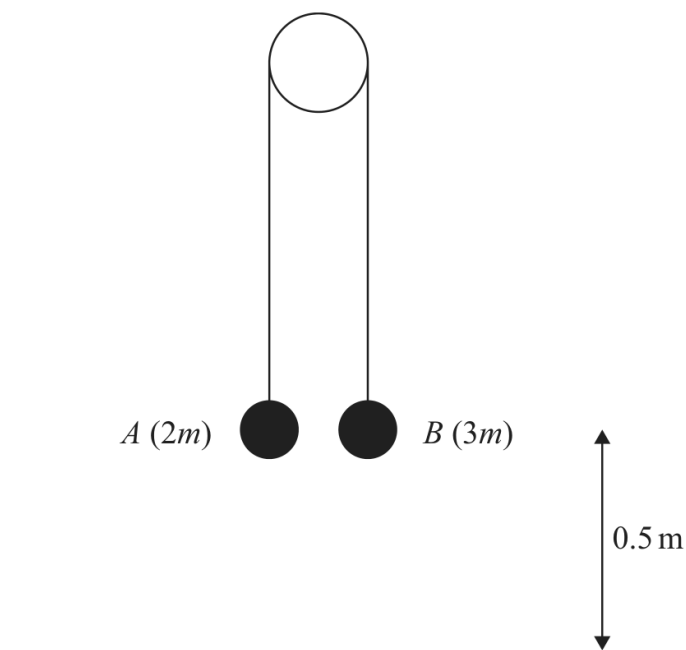
Initially, Q is 10 m above horizontal ground and P is more than 2 m below the pulley.

At the instant when Q has descended a distance of 2 m , the string breaks and Q falls to the ground.

(d) Find the speed of Q at the instant it hits the ground. (5)

Q23 PAST PAPER Pearson IAL WME01 · January 2025 · Q7**M1 · TOPIC 7** 15 marks

7.

**Figure 4**

One end of a light inextensible string is attached to a particle A of mass $2m$.
 The other end is attached to a particle B of mass $3m$.
 The string passes over a small smooth fixed pulley.
 The string is taut and both straight parts of the string are vertical.
 Both particles are held at a distance 0.5 m above a horizontal surface, as shown in Figure 4.

The system is released from rest and B moves downwards.

(a) Find the tension in the string in terms of m and g . (5)

(b) Find the speed of B at the instant it strikes the surface. (4)

In the subsequent motion, A does not reach the pulley and B does not rebound after it strikes the surface.

(c) Find the time from the instant when the system is released from rest to the instant when A first reaches a height of 1.06 m above the surface. (6)

Q24 PAST PAPER Pearson IAL WME01 · June 2025 · Q8

M1 · TOPIC 7 17 marks

8.

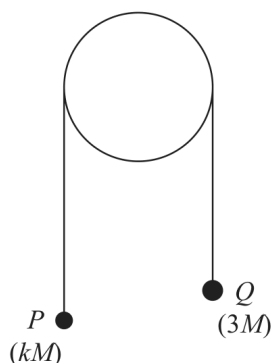


Figure 5

Two small balls, P and Q , have masses kM and $3M$ respectively, where $k < 3$

The balls are attached to the ends of a light inextensible string that passes over a fixed light smooth pulley.

The system is held at rest with the string taut and the hanging parts of the string vertical, as shown in Figure 5.

The system is released from rest and, in the subsequent motion, P moves with an acceleration of magnitude $\frac{1}{5}g$

The balls are modelled as particles.

(a) Write down an equation of motion for P .

(2)

(b) Find the value of k .

(3)

Given that $M = 0.5 \text{ kg}$,

(c) find the magnitude of the force exerted on the pulley by the string while Q is moving downwards.

(3)

At the instant when the system is released, P is more than 2.5 m from the pulley and Q is 2.5 m above horizontal ground.

After hitting the ground, Q rebounds with a speed of 0.4 ms^{-1}

(d) Find the magnitude of the impulse received by Q when it hits the ground.

(5)

In the subsequent motion, P does not hit the pulley.

(e) Find the total time from when the balls are released until P first comes to rest.

(4)

04 Pulleys with a Slope

Same method, with the slope rules from Topic 8: along the plane the weight gives $mg \sin \alpha$, perpendicular gives $R = mg \cos \alpha$, friction μR opposes the motion. One equation for the particle on the slope, one for the hanging particle.

WHEN YOU SEE → YOU DO

A on a rough slope, B hanging

$$\rightarrow T - mg \sin \alpha - \mu R = ma; Mg - T = Ma$$

"A comes to rest before the pulley" after B lands

→ new stage: A decelerates with weight and friction

"force exerted on the pulley by the string"

→ angle between the strings is $90^\circ - \alpha$ (vertical and slope): $2T \cos \frac{90^\circ - \alpha}{2}$

a horizontal string at the top (table on the other side)

→ angle between the strings $180^\circ - \alpha$

blocks added until it moves

→ limiting equilibrium, friction reverses

WORKED MODEL • OCTOBER 2021

A ($2m$) on a plane $\tan \alpha = \frac{3}{4}$, $\mu = \frac{1}{2}$; B ($3m$) hangs. Find a .

1 $R = \frac{8}{5}mg$, $F = \frac{4}{5}mg$

perpendicular, then μR

2 A: $T - \frac{6}{5}mg - \frac{4}{5}mg = 2ma$; B: $3mg - T = 3ma$

one equation each

ANSWER $mg = 5ma \Rightarrow a = \frac{g}{5}$

× **TRAP** Friction up the slope when A moves up. Friction always opposes the motion of A.

IDEA 04

Practise: Pulleys with a Slope

My questions first, then real past papers.

Q25 DR ESLAM Smooth slope and pulley

M1 · TOPIC 7 4 marks

A 4 kg particle on a smooth plane at 30° is joined over a pulley at the top to a 3 kg particle hanging freely. Find the acceleration.

Q26 DR ESLAM Rough slope and pulley

M1 · TOPIC 7 5 marks

The same system, but now $\mu = 0.2$ between the 4 kg particle and the plane. Find the acceleration.

Friction acts down the slope, against the motion.

PAST PAPERS

Q27 PAST PAPER Pearson IAL WME01 · January 2021 · Q8

M1 · TOPIC 7 17 marks

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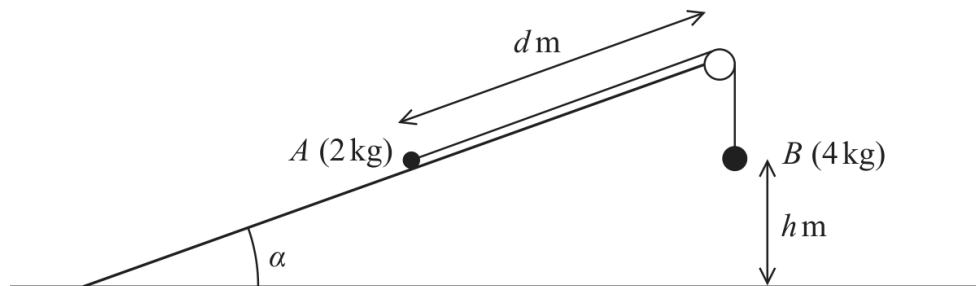


Figure 3

Two particles, A and B , have masses 2 kg and 4 kg respectively. The particles are connected by a light inextensible string. The string passes over a small smooth pulley which is fixed at the top of a rough plane. The plane is inclined to the horizontal ground at an angle α where $\tan \alpha = \frac{3}{4}$. The particle A is held at rest on the plane at a distance d metres from the pulley. The particle B hangs freely at rest, vertically below the pulley, at a distance h metres above the ground, as shown in Figure 3. The part of the string between A and the pulley is parallel to a line of greatest slope of the plane. The coefficient of friction between A and the plane is $\frac{1}{4}$.

The system is released from rest with the string taut and B descends.

- (a) Find the tension in the string as B descends. (9)

On hitting the ground, B immediately comes to rest.

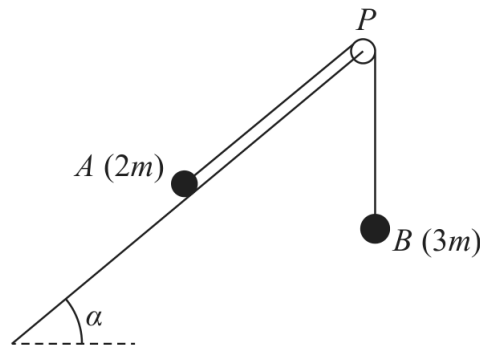
Given that A comes to rest before reaching the pulley,

- (b) find, in terms of h , the range of possible values of d . (7)
- (c) State one physical factor, other than air resistance, that could be taken into account to make the model described above more realistic. (1)

(Total 17 marks)

Q28 PAST PAPER Pearson IAL WME01 · October 2021 · Q7**M1 · TOPIC 7** 14 marks

7.

**Figure 4**

One end of a light inextensible string is attached to a particle A of mass $2m$. The other end of the string is attached to a particle B of mass $3m$. The string passes over a small, smooth, light pulley P which is fixed at the top of a rough inclined plane. The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$.

Particle A is held at rest on the plane with the string taut and B hanging freely below P , as shown in Figure 4. The section of the string AP is parallel to a line of greatest slope of the plane.

The coefficient of friction between A and the plane is $\frac{1}{2}$.

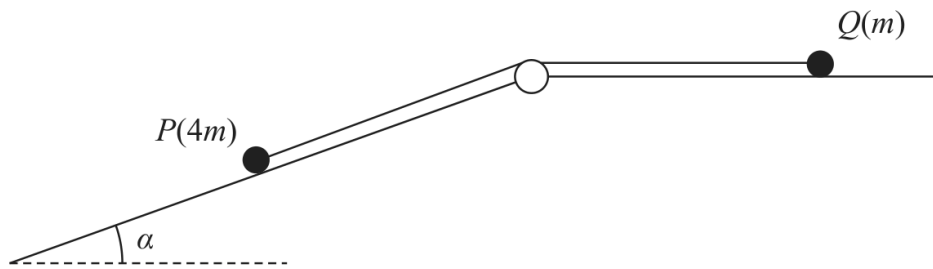
Particle A is released and begins to move up the plane.

For the motion before A reaches the pulley,

- (a) (i) write down an equation of motion for A ,
- (ii) write down an equation of motion for B ,
- (4)
- (b) find, in terms of g , the acceleration of A ,
- (5)
- (c) find the magnitude of the force exerted on the pulley by the string.
- (4)
- (d) State how you have used the information that P is a smooth pulley.
- (1)

Q29 PAST PAPER Pearson IAL WME01 · January 2022 · Q7**M1 · TOPIC 7** 13 marks

7.

**Figure 4**

A particle P of mass $4m$ lies on the surface of a fixed rough inclined plane.

The plane is inclined to the horizontal at an angle α where $\tan \alpha = \frac{3}{4}$

The particle P is attached to one end of a light inextensible string.

The string passes over a small smooth pulley that is fixed at the top of the plane. The other end of the string is attached to a particle Q of mass m which lies on a smooth horizontal plane.

The string lies along the horizontal plane and in the vertical plane that contains the pulley and a line of greatest slope of the inclined plane.

The system is released from rest with the string taut, as shown in Figure 4, and P moves down the plane.

The coefficient of friction between P and the plane is $\frac{1}{4}$

For the motion before Q reaches the pulley

(a) write down an equation of motion for Q , (1)

(b) find, in terms of m and g , the tension in the string, (7)

(c) find the magnitude of the force exerted on the pulley by the string. (4)

(d) State where in your working you have used the information that the string is light. (1)

Q30 PAST PAPER Pearson IAL WME01 · October 2023 · Q7**M1 · TOPIC 7** 13 marks

7.

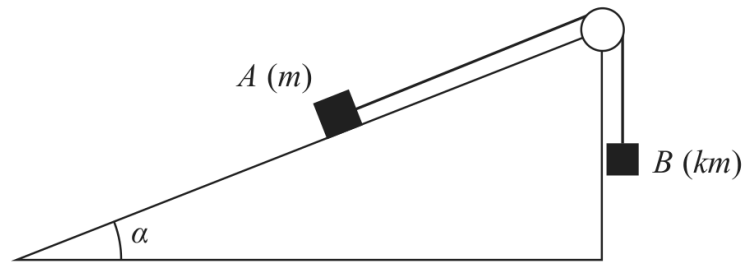
**Figure 4**

Figure 4 shows a block A of mass m held at rest on a rough plane. The plane is inclined at an angle α to the horizontal and the coefficient of friction between the block and the plane is μ .

One end of a light inextensible string is now attached to A . The string passes over a small smooth pulley which is fixed at the top of the plane. The other end of the string is attached to a block B of mass km . Block B hangs vertically below the pulley, with the string taut.

The string from A to the pulley lies along a line of greatest slope of the plane.

Both A and B are modelled as particles.

When the system is released from rest, A moves up the plane and the tension in the string is $\frac{4mg}{3}$

Given that $\mu = \frac{1}{3}$ and $\tan \alpha = \frac{7}{24}$

- (a) (i) find the magnitude of the acceleration of A , giving your answer in terms of g ,
 (ii) find the value of k .

(9)

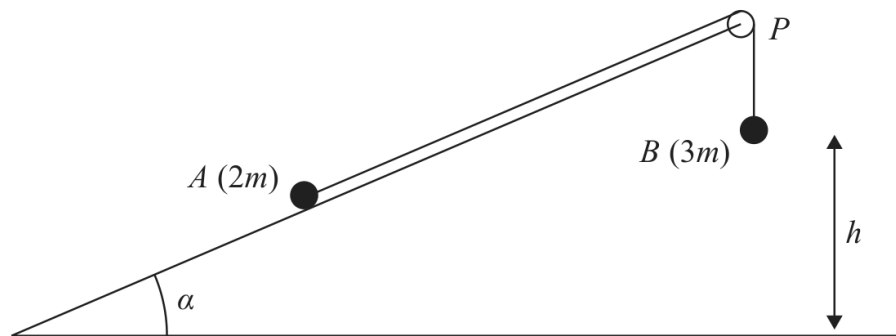
- (b) Find the magnitude of the resultant force exerted on the pulley by the string, giving your answer in terms of m and g .

(4)

Q31 PAST PAPER Pearson IAL WME01 · June 2024 · Q8

M1 · TOPIC 7 15 marks

8.

**Figure 4**

One end of a light inextensible string is attached to a particle A of mass $2m$. The other end of the string is attached to a particle B of mass $3m$. Particle A is held at rest on a rough plane which is inclined to horizontal ground at an angle α , where $\tan \alpha = \frac{5}{12}$.

The string passes over a small smooth pulley P which is fixed at the top of the plane. Particle B hangs vertically below P with the string taut, at a height h above the ground, as shown in Figure 4.

The part of the string between A and P lies along a line of greatest slope of the plane. The two particles, the string and the pulley all lie in the same vertical plane.

The coefficient of friction between A and the plane is $\frac{11}{36}$.

The particle A is released from rest and begins to move up the plane.

(a) Show that the frictional force acting on A as it moves up the plane is $\frac{22mg}{39}$ (3)

(b) Write down an equation of motion for B . (2)

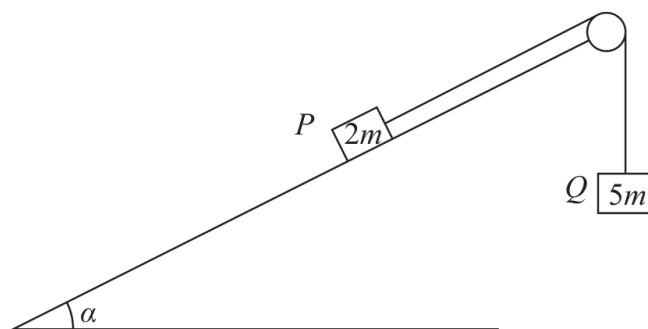
(c) Show that the acceleration of A immediately after its release is $\frac{1}{3}g$ (4)

In the subsequent motion, A comes to rest before it reaches the pulley.

(d) Find, in terms of h , the total distance travelled by A from when it was released from rest to when it first comes to rest again. (6)

Q32 PAST PAPER Pearson IAL WME01 · October 2025 · Q7**M1 · TOPIC 7** 14 marks

7.

**Figure 5**

A block P of mass $2m$ is held at rest on a fixed rough plane.

The plane is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{5}{12}$

One end of a light inextensible string is attached to P .

The string is parallel to a line of greatest slope of the plane and passes over a smooth light pulley which is fixed at the top of the plane.

The other end of the string is attached to a block Q of mass $5m$.

Block Q hangs vertically below the pulley, as shown in Figure 5.

The system is released from rest with the string taut and block P moves up the plane.

Immediately after the system is released, the tension in the string is T and the acceleration of the blocks is a .

The blocks are modelled as particles and air resistance is ignored.

(a) Write down an equation of motion for Q .

(2)

The coefficient of friction between P and the plane is $\frac{1}{8}$

(b) Find T in terms of m and g .

(7)

(c) State how the solution to part (b) uses the fact that the string is inextensible.

(1)

The magnitude of the force exerted on the pulley by the string is kmg .

(d) Find the value of k to 3 significant figures.

(4)

Q33 PAST PAPER Pearson IAL WME01 · January 2026 · Q7

M1 · TOPIC 7 14 marks

7.

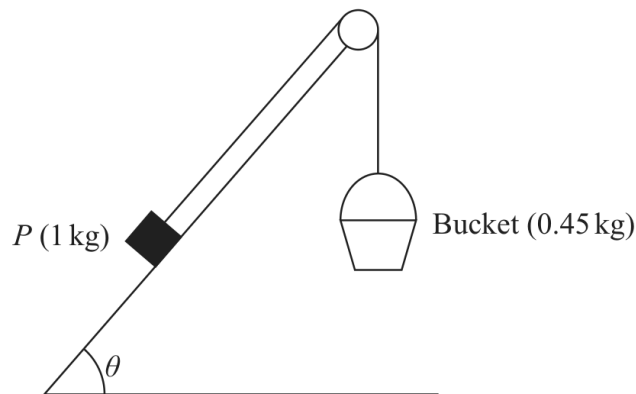


Figure 3

A rough plane is inclined at an angle θ to the horizontal, where $\tan \theta = \frac{4}{3}$

One end of a light inextensible string is attached to a package P .
 The string passes over a smooth pulley that is fixed at the top of the plane.
 The string from P to the pulley lies along a line of greatest slope of the plane.
 The other end of the string is attached to a bucket.
 The package P has mass 1 kg and is **held** at rest on the plane.
 The bucket has mass 0.45 kg and hangs vertically below the pulley with the string taut, as shown in Figure 3.

The coefficient of friction between P and the plane is μ

When a small block of mass 0.2 kg is placed in the bucket, the system is released and remains at rest in **limiting equilibrium**, with P on the point of slipping down the plane.

(a) Show that $\mu = \frac{1}{4}$ (7)

Additional identical blocks of mass 0.2 kg are added one at a time into the bucket until P starts to move **up** the plane.

At the instant when P starts to move up the plane, there are a **total** of n blocks in the bucket.

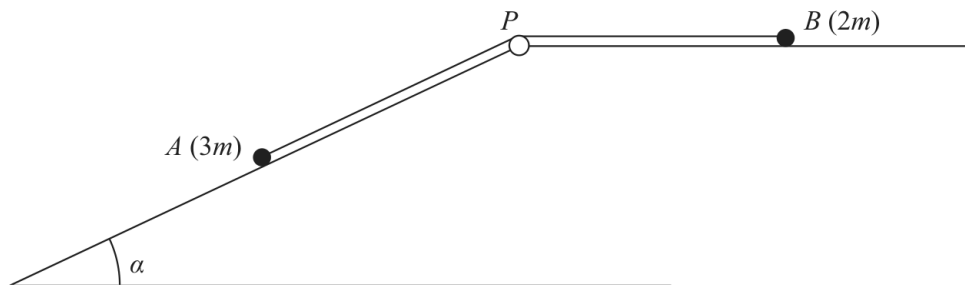
(b) Find (7)

- the value of n
- the magnitude of the initial acceleration of P

Q34 PAST PAPER Pearson IAL WME01 · June 2026 · Q8

M1 · TOPIC 7 14 marks

8.

**Figure 4**

One end of a light inextensible string is attached to a particle A of mass $3m$. The other end of the string is attached to a particle B of mass $2m$. Particle A lies on a **rough** plane which is inclined to the horizontal at an angle α , where $\tan \alpha = \frac{3}{4}$. Particle B lies on a **smooth** horizontal plane. The string passes over a small smooth pulley P which is fixed at the top of the inclined plane, as shown in Figure 4.

The two particles, the string and the pulley all lie in the same vertical plane. The portion of the string between A and P lies along a line of greatest slope of the inclined plane.

The coefficient of friction between A and the inclined plane is $\frac{1}{4}$.

The system is released from rest with the string taut and A begins to move down the inclined plane.

(a) Find, in terms of m and g , the magnitude of the frictional force acting on A . (3)

(b) Write down an equation of motion for B . (1)

(c) Show that the tension in the string is $\frac{12mg}{25}$. (4)

(d) State how you have used the fact that the string is inextensible. (1)

The force exerted on the pulley by the string is S .

(e) State clearly, giving your answer in terms of α , the direction of S . (1)

(f) Find, in terms of m and g , the magnitude of S . (4)

(Total for Question 8 is 14 marks)

ALL The Whole Topic on One Page

Tear this page out before the exam.

- | | | |
|-----------|-------------------------|--|
| 01 | One body | With – against = ma , in the direction of a . |
| 02 | In a line | Whole system for a ; one part for the tension. |
| 03 | Pulleys | One equation each; add to lose T ; pulley force from two tensions. |
| 04 | Slope and pulley | $mg \sin \alpha$, $\mu mg \cos \alpha$ against the motion. |

BOTTOM LINE Diagram → direction of a → one equation per particle → add → SUVAT.

Answers

Newton's Second Law · for the teacher. Every answer checked with a computer algebra system.

IDEA 01 · ONE BODY: RESULTANT FORCE = ma

- Q1** Dr Eslam · Driving force and resistance $2400 = 1200a \Rightarrow a = 2 \text{ m s}^{-2}$
- Q2** Dr Eslam · A lift $T = 560(10.3) = 5768 \text{ N}; R = 60(10.3) = 618 \text{ N}$
- Q3** Dr Eslam · Falling with resistance $0.5g - 1.4 = 0.5a \Rightarrow a = 7; v = 21 \text{ m s}^{-1}$
- Q4** Pearson IAL WME01 · June 2021 · Q4 (a) $T = 400 \text{ N}$ (b) $R \approx 119 \text{ N}$
- Q5** Pearson IAL WME01 · October 2021 · Q2 (a) u (b) reversed (c) $\mu = \frac{1}{3}$
- Q6** Pearson IAL WME01 · October 2022 · Q4 (a) 1.5 m s^{-2} (b) 50 kg
- Q7** Pearson IAL WME01 · June 2023 · Q3 (a) 0.553 s (b) $R \approx 0.293 \text{ N}$
- Q8** Pearson IAL WME01 · October 2023 · Q3 (a) $v = 9$ (b) 1.8 N s (c) $R \approx 695 \text{ N}$
- Q9** Pearson IAL WME01 · January 2025 · Q3 (a) $\lambda = 15m$ (b) $\mu \approx 0.68$ (c) 1.8 m s^{-1}
- Q10** Pearson IAL WME01 · June 2025 · Q4 (a) $\alpha \approx 43^\circ$ (b) 0.023 m s^{-2}

IDEA 02 · CONNECTED IN A LINE: TOWING AND LIFTS

- Q11** Dr Eslam · Car and trailer $a = 1.3 \text{ m s}^{-2}; T - 150 = 650 \Rightarrow T = 800 \text{ N}$
- Q12** Dr Eslam · Two particles on one string $a = \frac{60-5g}{5} = 2.2; T = 3(9.8 + 2.2) = 36 \text{ N}$
- Q13** Pearson IAL WME01 · October 2019 · Q3 $D = 6000$
- Q14** Pearson IAL WME01 · June 2022 · Q3 (a) $R = 5600$ (b) the tractor and the block have the same acceleration
- Q15** Pearson IAL WME01 · January 2023 · Q7 (a) $5m$ (b) $2m$
- Q16** Pearson IAL WME01 · June 2023 · Q7 (a) $R = 60$ (b) 1000 N (c) 78.5 m
- Q17** Pearson IAL WME01 · June 2024 · Q3 $T = \frac{9}{2}mg$

IDEA 03 · PULLEYS: HANGING AND ON A TABLE

- Q18** Dr Eslam • Two hanging masses
 $a = \frac{2g}{8} = 2.45 \text{ m s}^{-2}; T = 36.75 \text{ N}$
- Q19** Dr Eslam • Table and edge
 $a = \frac{g}{3} \approx 3.27; T \approx 6.53 \text{ N}; \text{ pulley: } T\sqrt{2} \approx 9.24 \text{ N}$
- Q20** Pearson IAL WME01 • January 2020 • Q7
 (a) $1.4 = \frac{1}{2}a(2^2) \Rightarrow a = 0.7$ (b) the string is inextensible (c) 38.6 N (d) $\mu = 0.625$ (e) A does not reach the pulley
- Q21** Pearson IAL WME01 • October 2022 • Q7
 (a) $k = \frac{1+\mu}{2}$ (b) $t = \sqrt{\frac{4d}{g(1-\mu)}}$ (c) the system does not move
- Q22** Pearson IAL WME01 • October 2024 • Q5
 (a) $T - 3g = 3a$ (b) $T = \frac{15g}{4} \approx 36.8 \text{ N}$ (c) $F = \frac{15g}{2} = 73.5 \text{ N}$ (d) $\sqrt{17g} \approx 12.9 \text{ m s}^{-1}$
- Q23** Pearson IAL WME01 • January 2025 • Q7
 (a) $T = \frac{12mg}{5}$ (b) 1.4 m s^{-1} (c) 0.767 s
- Q24** Pearson IAL WME01 • June 2025 • Q8
 (a) $T - kMg = \frac{1}{5}kMg$ (b) $k = 2$ (c) $\frac{12g}{5} \approx 23.5 \text{ N}$ (d) $1.5(\sqrt{g} + 0.4) \approx 5.30 \text{ N s}$ (e) $\frac{6}{\sqrt{g}} \approx 1.92 \text{ s}$

IDEA 04 • PULLEYS WITH A SLOPE

- Q25** Dr Eslam • Smooth slope and pulley
 $3g - 4g \sin 30^\circ = 7a \Rightarrow a = 1.4 \text{ m s}^{-2}$
- Q26** Dr Eslam • Rough slope and pulley
 $F = 0.2(4g \cos 30^\circ) \approx 6.79$; $a = \frac{29.4 - 19.6 - 6.79}{7} \approx 0.43 \text{ m s}^{-2}$
- Q27** Pearson IAL WME01 • January 2021 • Q8
 (a) $T = \frac{12g}{5} \approx 23.5 \text{ N}$ (b) $d > \frac{3h}{2}$ (c) e.g. friction at the pulley
- Q28** Pearson IAL WME01 • October 2021 • Q7
 (a)(i) $T - 2mg \sin \alpha - F = 2ma$ (ii) $3mg - T = 3ma$
 (b) $a = \frac{g}{5}$ (c) $\frac{48\sqrt{5}}{25}mg$ (d) same tension on both sides
- Q29** Pearson IAL WME01 • January 2022 • Q7
 (a) $T = ma$ (b) $T = \frac{8mg}{25}$ (c) $\frac{8\sqrt{10}}{125}mg$ (d) the tension is the same throughout
- Q30** Pearson IAL WME01 • October 2023 • Q7
 (a)(i) $\frac{11g}{15}$ (ii) $k = 5$ (b) $\frac{32mg}{15}$
- Q31** Pearson IAL WME01 • June 2024 • Q8
 (a) $F = \frac{11}{36} \cdot \frac{12}{13}(2mg) = \frac{22mg}{39}$ (b) $3mg - T = 3ma$
 (c) $a = \frac{g}{3}$ (d) $\frac{3h}{2}$
- Q32** Pearson IAL WME01 • October 2025 • Q7
 (a) $5mg - T = 5ma$ (b) $a = \frac{4g}{7}$, $T = \frac{15mg}{7}$ (c) same magnitude of acceleration (d) $k \approx 3.57$
- Q33** Pearson IAL WME01 • January 2026 • Q7
 (a) $\mu = \frac{1}{4}$ (b)(i) $n = 3$ (ii) $\frac{2g}{41} \approx 0.478 \text{ m s}^{-2}$
- Q34** Pearson IAL WME01 • June 2026 • Q8
 (a) $\frac{3mg}{5}$ (b) $T = 2ma$ (c) $T = \frac{12mg}{25}$ (d) same magnitude of acceleration (e) below the horizontal at $90^\circ - \frac{\alpha}{2}$ to it (f) $\frac{12\sqrt{10}}{125}mg$